

Trend-Aware Cold-Start Recommendation via Multimodal Similarity Boosting

Jacob Simmons

jss0112@auburn.edu

Owen Strength

ods0005@auburn.edu

Abstract

Cold-start items on short-video platforms receive minimal exposure because collaborative filtering signals are absent at upload time. We propose a modular trend-aware boosting mechanism that uses multimodal features—visual, textual, and acoustic—from the MicroLens dataset to estimate a new video’s semantic similarity to currently trending content. Rather than building a rigid end-to-end pipeline, our approach acts as a pluggable score-adjustment layer on top of an existing recommender. A personalization constraint will scale the boost based on each user’s historical affinity for viral content, protecting users who prefer niche material. We plan to evaluate on MicroLens with standard ranking metrics and cold-start-specific data splits.

1. Introduction

Short-video platforms such as TikTok, YouTube Shorts, and Instagram Reels face a persistent challenge: newly uploaded videos lack the interaction history that collaborative filtering models depend on. This *cold-start problem* means that fresh content is systematically under-recommended, reducing creator satisfaction and platform diversity [5]. While prior work has explored content-based and hybrid solutions, many of these approaches require training complex end-to-end models that are difficult to integrate into existing recommendation stacks [8].

We observe that trending content provides a useful distributional signal. Videos that are currently gaining traction occupy a recognizable region of multimodal feature space, and a cold item’s proximity to that region may be a lightweight predictor of its relevance. Our core idea is to compute a *trending boost*—a score adjustment derived from the multimodal similarity between a new video and recent trends—and inject it into a base recommender’s output.

A naive uniform boost, however, could harm users who prefer niche or long-tail content. We therefore propose a *personalization constraint* that modulates the boost according to each user’s historical engagement with viral material. The goal is to surface promising cold items to trend-receptive users without disrupting the experience of others.

2. Problem Statement

We study the cold-start item recommendation problem in the short-video domain, using the MicroLens dataset [4]. Cold-start items are defined as videos with limited or no interactions (below 10) at recommendation time; we will experiment with different thresholds to evaluate robustness across different levels of sparsity. Our objective is to improve the ranking of these items without degrading overall recommendation quality.

Proposed approach. At a high level, our method consists of three components: (1) a *trend identification* step that selects currently popular videos using a sliding time window over the interaction logs, (2) a *similarity scoring* step that measures how close each cold video’s multimodal embedding is to the trend cluster, and (3) a *personalized scaling* step that adjusts the resulting boost based on per-user virality affinity. The exact formulation of each component—including the choice of similarity metric, trend-window size, aggregation strategy, and affinity definition—will be explored and refined during the project.

3. Data

We will use the MicroLens dataset [4], a large-scale benchmark for short-video recommendation that includes pre-extracted multimodal features (visual, textual, and audio embeddings) alongside timestamped user-item interaction logs. Multiple dataset scales are available (e.g., 50K and 100K user subsets); we will begin with the smaller split for rapid prototyping and scale up as needed. No additional data collection is required.

4. Related Work and Background Reading

Our reading so far spans several relevant areas, and we expect this list to grow as the project progresses.

Short-video user modeling. He et al. [3] study segment-level interest modeling for short-video recommendation, proposing a multimodal encoder and segment interest decoder that captures how user preferences shift *within* a single video. Their work highlights the importance of fine-grained temporal modeling in the short-video domain and

motivates our own use of multimodal signals to characterize video content.

Multimodal encoding for recommendation. UniRec [6] introduces a unified multimodal encoder for LLM-based recommendation that handles text, images, categorical, and numerical features through a schema-aware triplet representation and hierarchical Q-Former. This work demonstrates the value of principled multimodal fusion for recommendation and informs our thinking about how to combine heterogeneous video embeddings.

Cold-start and content-based methods. DropoutNet [7] addresses cold start by training a deep model to be robust to missing collaborative signals, learning to fall back on content features when interaction data is absent. VBPR [2] integrates visual features into Bayesian personalized ranking, showing that visual similarity can meaningfully improve item recommendation even without rich interaction histories. Both inform our strategy of leveraging multimodal content to compensate for sparse feedback on new items.

Popularity bias and trend dynamics. Abdollahpouri et al. [1] analyze how popularity bias in recommender systems leads to unfair exposure of long-tail items, motivating our personalization constraint that prevents trend-receptive boosting from overwhelming niche-preferring users. We also plan to review temporal popularity modeling literature to guide our trending-window design.

The MicroLens benchmark paper [4] will serve as our primary reference for dataset-specific baselines and evaluation protocols.

5. Evaluation Plan

Hypotheses. We test three claims: (H1) trend-aware boosting significantly improves $\text{NDCG}@K$ for cold-start items relative to a base recommender with no boost; (H2) the personalization constraint prevents degradation in recommendation quality for users whose histories skew toward long-tail content; and (H3) similarity-based boosting outperforms a popularity-only re-ranking heuristic by leveraging multimodal content signals rather than raw interaction counts.

Quantitative metrics. We plan to report standard ranking metrics such as $\text{Hit Rate}@K$ and $\text{NDCG}@K$ across multiple values of K . Results will be broken down by item type (cold-start vs. warm) to verify that our boost helps new items without hurting overall quality. We will compare against at least: (i) the base recommender without any boost, (ii) a popularity-based heuristic, and (iii) a content-only nearest-neighbor baseline. We intend to run experiments over multiple random seeds and assess significance.

Qualitative analysis. We expect to produce visualizations such as embedding-space plots (e.g., t -SNE) showing how cold items relate to trending clusters, and performance curves stratified by user virality-affinity quantiles. An abla-

tion removing the personalization constraint will help isolate its contribution.

6. Timeline

Weeks 1–2: Data exploration, feature analysis, and baseline model training.

Weeks 3–4: Implement trend identification and similarity-based boosting module.

Weeks 5–6: Implement personalized scaling; integrate all components and iterate on design choices.

Week 7: Run evaluations, ablations, and generate figures.

Week 8: Write final report and prepare presentation.

References

- [1] Himan Abdollahpouri, Masoud Mansoury, Robin Burke, and Bamshad Mobasher. The unfairness of popularity bias in recommendation. In *RecSys Workshop on Recommendation in Multistakeholder Environments*, 2019. 2
- [2] Ruining He and Julian McAuley. VBPR: Visual Bayesian personalized ranking from implicit feedback. In *Proceedings of the AAAI Conference on Artificial Intelligence*, 2016. 2
- [3] Zhiyu He, Zhixin Ling, Jiayu Li, Zhiqiang Guo, Weizhi Ma, Xinchun Luo, Min Zhang, and Guorui Zhou. Short video segment-level user dynamic interests modeling in personalized recommendation. In *Proceedings of the 48th International ACM SIGIR Conference on Research and Development in Information Retrieval*, 2025. 1
- [4] Yongxin Ni, Yu Cheng, Xiangyan Liu, Junchen Fu, Youhua Li, Xiangnan He, Yongfeng Zhang, and Fajie Yuan. MicroLens: A content-driven micro-video recommendation dataset at scale. In *Advances in Neural Information Processing Systems (NeurIPS) Datasets and Benchmarks Track*, 2023. 1, 2
- [5] Andrew I. Schein, Alexandrin Popescul, Lyle H. Ungar, and David M. Pennock. Methods and metrics for cold-start recommendations. In *Proceedings of the 25th Annual International ACM SIGIR Conference on Research and Development in Information Retrieval*, pages 253–260. ACM, 2002. 1
- [6] UniRec Authors. UniRec: Unified multimodal encoding for LLM-based recommendations. *arXiv preprint arXiv:2601.19423*, 2025. 2
- [7] Maksims Volkovs, Guangwei Yu, and Tomi Poutanen. DropoutNet: Addressing cold start in recommender systems. In *Advances in Neural Information Processing Systems*, 2017. 2
- [8] Yongchun Zhu, Kaikai Lin, Peilin Zhao, et al. Addressing the item cold-start problem by attribute-driven active learning. *IEEE Transactions on Knowledge and Data Engineering*, 32(4):631–644, 2020. 1